

Students:

Aryam Aldayhani
431215431

Alanoud Almazrua
431201829

Razan Alomari
432205459

Jory Alghofaily
431201878

Mayar Aldubayyan
431201869

How it supports vision 2030?



Rasid advances Saudi Vision 2030 "Digital Transformation" goals by establishing a tamper-proof, blockchain-integrated infrastructure for vehicular lifecycle management. By eliminating odometer fraud and information asymmetry through automated IoT validation, the system restores market trust and supports "Smart Mobility".



Introduction

Rasid is a Digital Vehicle Passport (DVP) system designed to combat fraud like odometer tampering and hidden accident records. By integrating IoT and Blockchain, it creates a secure, decentralized ledger that ensures absolute integrity throughout a vehicle's lifecycle. This transformation provides a transparent, trusted environment, ensuring every stage of a vehicle's history is documented and verifiable.



Objectives



System Integration: Seamless integration of IoT and Blockchain for secure vehicle data recording.



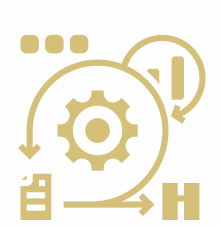
Data Protection: Advanced encryption and access control to protect sensitive information.



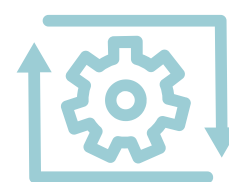
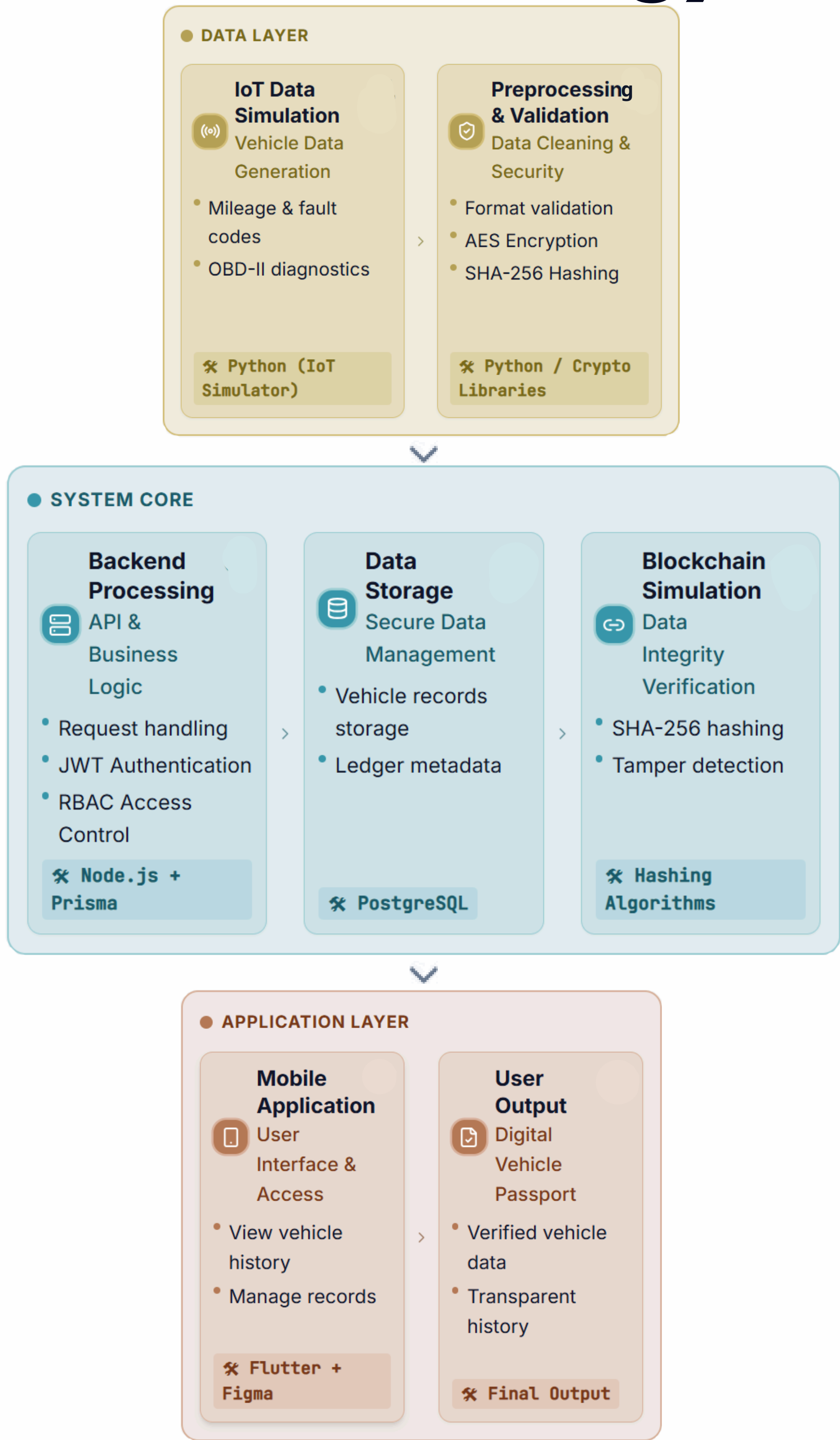
Lifecycle Management: Tracking the complete digital lifecycle of vehicles from production to disposal.



Fraud Reduction: Enhancing transparency to minimize fraud and financial risks.



Methodology



Experiments

ARCHITECTURE
COMPARISON

PUBLIC BLOCKCHAIN		
Latency	~15 seconds	
Cost	High (Gas Fees)	
Security	Tamper-proof	
IOT Integration	Complex	

VS

HYBRID LEDGER (RASID)		
Latency	< 1 second	
Cost	Zero	
Security	SHA-256 tamper-proof	
IoT Integration	Seamless & scalable	

Proposed solution:

The Hybrid Ledger architecture was adopted, which helped us in:

- Achieving real-time IoT data processing
- Zero transaction costs
- maintaining strict tamper-evident security.

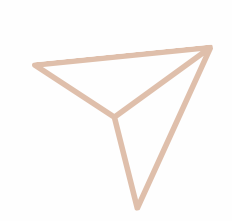


Prototype



Conclusion

Rasid secures the automotive market by merging IoT and Blockchain into a "Digital Vehicle Passport". The system eliminates fraud and data fragmentation, ensuring total transparency. By providing a trusted, immutable record, it protects economic value and directly supports the digital goals of Saudi Vision.



Future Work

- Government Integration:** linking with national traffic systems (Absher, Mojaz) for real-time vehicle data.
- IoT & Blockchain Infrastructure:** deploying real IoT devices connected to a secure blockchain-based system.
- AI Predictive Maintenance:** using AI to predict failures and reduce maintenance costs.

Department of Information Technology

Supervisor: Dr. Eman Abouelkheir



Tools



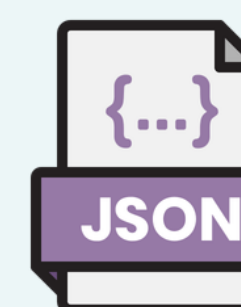
Flutter



Prisma



TimescaleDB



JSON



Python



PostgreSQL



Node.js



Express.js



SHA-256